2.39 Integration: Double Angles

Worked Solutions 1–16

1 a)

$\cos 2x = 2\cos^2 x - 1$

1 b)

$$\begin{aligned}
\cos 2x = 2\cos^2 x - 1 &\Rightarrow \cos^2 x = \tfrac{1}{2}(\cos 2x + 1) \\
&\Rightarrow \cos^2 x = \tfrac{1}{2}\cos 2x + \tfrac{1}{2} \\
\int_0^{\pi/4} \cos^2 x\, dx &= \int_0^{\pi/4} (\tfrac{1}{2}\cos 2x + \tfrac{1}{2})\, dx \\
&= [\tfrac{1}{4}\sin 2x + \tfrac{1}{2}x]_0^{\pi/4} \\
&= (\tfrac{1}{4}\sin(2\times\tfrac{\pi}{4}) + \tfrac{1}{2}\times\tfrac{\pi}{4}) - (\tfrac{1}{4}\sin(2\times 0) + \tfrac{1}{2}\times 0) \\
&= (\tfrac{1}{4} + \tfrac{\pi}{8}) - (0+0) = \tfrac{2}{8} + \tfrac{\pi}{8} = \tfrac{2+\pi}{8}
\end{aligned}$$

2 a)

$\cos 2x = 1 - 2\sin^2 x$

2 b)

$$\begin{aligned}
\cos 2x = 1 - 2\sin^2 x &\Rightarrow \sin^2 x = \tfrac{1}{2}(1 - \cos 2x) \\
&\Rightarrow \sin^2 x = \tfrac{1}{2} - \tfrac{1}{2}\cos 2x \\
\int_0^{\pi/2} \sin^2 x\, dx &= \int_0^{\pi/2} (\tfrac{1}{2} - \tfrac{1}{2}\cos 2x)\, dx \\
&= [\tfrac{1}{2}x - \tfrac{1}{4}\sin 2x]_0^{\pi/2} \\
&= (\tfrac{1}{2}\times\tfrac{\pi}{2} - \tfrac{1}{4}\sin(2\times\tfrac{\pi}{2})) - (\tfrac{1}{2}\times 0 - \tfrac{1}{4}\sin(2\times 0)) \\
&= (\tfrac{\pi}{4} - 0) - (0-0) = \tfrac{\pi}{4}
\end{aligned}$$

3 a)

$$\begin{aligned}
\int (1+\sin x)^2\, dx &= \int (1 + 2\sin x + \sin^2 x)\, dx \\
&= \int \left(1 + 2\sin x + (\tfrac{1}{2} - \tfrac{1}{2}\cos 2x)\right) dx \\
&= \int (\tfrac{3}{2} + 2\sin x - \tfrac{1}{2}\cos 2x)\, dx \\
&= \tfrac{3}{2}x - 2\cos x - \tfrac{1}{4}\sin 2x + c
\end{aligned}$$

3 b)

$$\begin{aligned}\int (1-\cos x)^2\,dx &= \int (1-2\cos x+\cos^2 x)\,dx \\ &= \int (1-2\cos x+(\tfrac{1}{2}\cos 2x+\tfrac{1}{2}))\,dx \\ &= \int (\tfrac{3}{2}-2\cos x+\tfrac{1}{2}\cos 2x)\,dx \\ &= \tfrac{3}{2}x-2\sin x+\tfrac{1}{4}\sin 2x+c\end{aligned}$$

3 c)

$$\begin{aligned}\int (\sin x-\cos x)^2\,dx &= \int (\sin^2 x-2\sin x\cos x+\cos^2 x)\,dx \\ &= \int (\sin^2 x+\cos^2 x-2\sin x\cos x)\,dx \\ &= \int (1-\sin 2x)\,dx = x+\tfrac{1}{2}\cos 2x+c\end{aligned}$$

3 d)

$$\begin{aligned}\int (\sin x-\cos x)(\sin x+\cos x)\,dx &= \int (\sin^2 x-\cos^2 x)\,dx \\ &= \int ((\tfrac{1}{2}-\tfrac{1}{2}\cos 2x)-(\tfrac{1}{2}\cos 2x+\tfrac{1}{2}))\,dx \\ &= \int (\tfrac{1}{2}-\tfrac{1}{2}\cos 2x-\tfrac{1}{2}\cos 2x-\tfrac{1}{2})\,dx \\ &= \int (-\cos 2x)\,dx = -\tfrac{1}{2}\sin 2x+c\end{aligned}$$

4 a)

$$\int \sin x\cos x\,dx = \int \tfrac{1}{2}(2\sin x\cos x)\,dx = \int \tfrac{1}{2}\sin 2x\,dx = -\tfrac{1}{4}\cos 2x+c$$

4 b)

$$\int 8\sin 2x\cos 2x\,dx = \int 4(2\sin 2x\cos 2x)\,dx = \int 4\sin 4x\,dx = -\cos 4x+c$$

4 c)

$$\int 12\sin 5x\cos 5x\,dx = \int 6(2\sin 5x\cos 5x)\,dx = \int 6\sin 10x\,dx = -\tfrac{3}{5}\cos 10x+c$$

4 d)

$$\begin{aligned}\int \cos^2 6x \tan 6x \, dx &= \int \cos^2 6x \times \frac{\sin 6x}{\cos 6x} \, dx \\ &= \int \sin 6x \cos 6x \, dx = \int \tfrac{1}{2}(2 \sin 6x \cos 6x) \, dx \\ &= \int \tfrac{1}{2} \sin 12x \, dx = -\tfrac{1}{24} \cos 12x + c\end{aligned}$$

5 a)

$$\begin{aligned}\int_0^{\pi/2} (4 - 3\sin x)^2 \, dx &= \int_0^{\pi/2} (16 - 24 \sin x + 9 \sin^2 x) \, dx \\ &= \int_0^{\pi/2} \left(16 - 24 \sin x + 9(\tfrac{1}{2} - \tfrac{1}{2} \cos 2x)\right) dx \\ &= \int_0^{\pi/2} (16 - 24 \sin x + \tfrac{9}{2} - \tfrac{9}{2} \cos 2x) \, dx \\ &= \int_0^{\pi/2} (\tfrac{41}{2} - 24 \sin x - \tfrac{9}{2} \cos 2x) \, dx \\ &= [\tfrac{41}{2} x + 24 \cos x - \tfrac{9}{4} \sin 2x]_0^{\pi/2} \\ &= (\tfrac{41}{2} \times \tfrac{\pi}{2} + 24 \cos \tfrac{\pi}{2} - \tfrac{9}{4} \sin(2 \times \tfrac{\pi}{2})) \\ &\quad - (\tfrac{41}{2} \times 0 + 24 \cos 0 - \tfrac{9}{4} \sin(2 \times 0)) \\ &= (\tfrac{41\pi}{4} + 0 - 0) - (0 + 24 - 0) = \tfrac{41\pi}{4} - 24\end{aligned}$$

5 b)

$$\begin{aligned}\int_0^{\pi/2} (1 - 6\cos x)^2 \, dx &= \int_0^{\pi/2} (1 - 12 \cos x + 36 \cos^2 x) \, dx \\ &= \int_0^{\pi/2} \left(1 - 12 \cos x + 36(\tfrac{1}{2} \cos 2x + \tfrac{1}{2})\right) dx \\ &= \int_0^{\pi/2} (1 - 12 \cos x + 18 \cos 2x + 18) \, dx \\ &= \int_0^{\pi/2} (19 - 12 \cos x + 18 \cos 2x) \, dx \\ &= [19x - 12 \sin x + 9 \sin 2x]_0^{\pi/2} \\ &= (19 \times \tfrac{\pi}{2} - 12 \sin \tfrac{\pi}{2} + 9 \sin(2 \times \tfrac{\pi}{2})) \\ &\quad - (19 \times 0 - 12 \sin 0 + 9 \sin(2 \times 0)) \\ &= (\tfrac{19\pi}{2} - 12 + 0) - (0 - 0 + 0) = \tfrac{19\pi}{2} - 12\end{aligned}$$

6 a)

$$\begin{aligned}\int_0^{\pi/4} (1 - 8\sin 2x)^2 \, dx &= \int_0^{\pi/4} (1 - 16 \sin 2x + 64 \sin^2 2x) \, dx \\ &= \int_0^{\pi/4} \left(1 - 16 \sin 2x + 64(\tfrac{1}{2} - \tfrac{1}{2} \cos 4x)\right) dx\end{aligned}$$

$$
\begin{aligned}
&= \int_0^{\pi/4} (1 - 16\sin 2x + 32 - 32\cos 4x)\,dx \\
&= \int_0^{\pi/4} (33 - 16\sin 2x - 32\cos 4x)\,dx \\
&= [33x + 8\cos 2x - 8\sin 4x]_0^{\pi/4} \\
&= (33 \times \tfrac{\pi}{4} + 8\cos(2 \times \tfrac{\pi}{4}) - 8\sin(4 \times \tfrac{\pi}{4})) \\
&\quad - (33 \times 0 + 8\cos(2 \times 0) - 8\sin(4 \times 0)) \\
&= (\tfrac{33\pi}{4} + 0 - 0) - (0 + 8 - 0) = \tfrac{33\pi}{4} - 8
\end{aligned}
$$

6 b)

$$
\begin{aligned}
\int_{-\pi}^{\pi} (7 + 9\cos 3x)^2\,dx &= \int_{-\pi}^{\pi} (49 + 126\cos 3x + 81\cos^2 3x)\,dx \\
&= \int_{-\pi}^{\pi} (49 + 126\cos 3x + 81(\tfrac{1}{2}\cos 6x + \tfrac{1}{2}))\,dx \\
&= \int_{-\pi}^{\pi} (49 + 126\cos 3x + \tfrac{81}{2}\cos 6x + \tfrac{81}{2})\,dx \\
&= \int_{-\pi}^{\pi} (\tfrac{179}{2} + 126\cos 3x + \tfrac{81}{2}\cos 6x)\,dx \\
&= [\tfrac{179}{2}x + 42\sin 3x + \tfrac{27}{4}\sin 6x]_{-\pi}^{\pi} \\
&= (\tfrac{179}{2} \times \pi + 42\sin(3 \times \pi) + \tfrac{27}{4}\sin(6 \times \pi)) \\
&\quad - (\tfrac{179}{2} \times (-\pi) + 42\sin(3 \times (-\pi)) + \tfrac{27}{4}\sin(6 \times (-\pi))) \\
&= (\tfrac{179\pi}{2} + 0 + 0) - (-\tfrac{179\pi}{2} + 0 + 0) = 179\pi
\end{aligned}
$$

7 a)

$\int (2\cos^2 3x - 1)\,dx = \int \cos 6x\,dx = \frac{1}{6}\sin 6x + c$

7 b)

$\int (1 - 2\sin^2 5x)\,dx = \int \cos 10x\,dx = \frac{1}{10}\sin 10x + c$

7 c)

$$
\begin{aligned}
\int (\sin 4x \cot 2x - 1)\,dx &= \int \left(2\sin 2x\cos 2x \times \frac{\cos 2x}{\sin 2x} - 1\right) dx \\
&= \int (2\cos^2 2x - 1)\,dx = \int \cos 4x\,dx = \tfrac{1}{4}\sin 4x + c
\end{aligned}
$$

7 d)

$$
\begin{aligned}
\int (1 - \sin 20x \tan 10x)\,dx &= \int \left(1 - 2\sin 10x\cos 10x \times \frac{\sin 10x}{\cos 10x}\right) dx \\
&= \int (1 - 2\sin^2 10x)\,dx = \int \cos 20x\,dx = \tfrac{1}{20}\sin 20x + c
\end{aligned}
$$

8)

$$\int \frac{10\sin x\cos x}{2\cos^2 x+3}\,dx = \int \frac{5(2\sin x\cos x)}{(2\cos^2 x-1)+4}\,dx$$
$$= \int \frac{5\sin 2x}{\cos 2x+4}\,dx = -\tfrac{5}{2}\ln|\cos 2x+4| + c$$

9)

$$\frac{dy}{dx} = \frac{4\tan x}{\sqrt{2\cos 2x+2}} \Rightarrow \frac{dy}{dx} = \frac{4\tan x}{\sqrt{2(2\cos^2 x-1)+2}}$$
$$\Rightarrow \frac{dy}{dx} = \frac{4\tan x}{\sqrt{4\cos^2 x-2+2}}$$
$$\Rightarrow \frac{dy}{dx} = \frac{4\tan x}{\sqrt{4\cos^2 x}} \Rightarrow \frac{dy}{dx} = \frac{4\tan x}{2\cos x}$$
$$\Rightarrow \frac{dy}{dx} = 2\tan x\sec x \Rightarrow y = 2\sec x + c$$

When $x=\pi$, $y=6$:

$$6 = 2\sec\pi + c \Rightarrow 6 = \frac{2}{\cos\pi} + c \Rightarrow 6 = -2 + c \Rightarrow c = 8$$

Therefore $y = 2\sec x + 8$.

10)

$$\int_0^{\pi/4} \frac{\sqrt{1-\cos 2x}}{\cos 2x+1}\,dx = \int_0^{\pi/4} \frac{\sqrt{1-(1-2\sin^2 x)}}{(2\cos^2 x-1)+1}\,dx$$
$$= \int_0^{\pi/4} \frac{\sqrt{2\sin^2 x}}{2\cos^2 x}\,dx = \int_0^{\pi/4} \frac{\sqrt{2}\sin x}{2\cos^2 x}\,dx$$
$$= \int_0^{\pi/4} \tfrac{\sqrt{2}}{2}\tan x\sec x\,dx = [\tfrac{\sqrt{2}}{2}\sec x]_0^{\pi/4}$$
$$= (\tfrac{\sqrt{2}}{2}\sec\tfrac{\pi}{4}) - (\tfrac{\sqrt{2}}{2}\sec 0)$$
$$= (\tfrac{\sqrt{2}}{2}\times\tfrac{1}{\cos(\pi/4)}) - (\tfrac{\sqrt{2}}{2}\times\tfrac{1}{\cos 0})$$
$$= (\tfrac{\sqrt{2}}{2}\times\sqrt{2}) - (\tfrac{\sqrt{2}}{2}\times 1) = 1 - \tfrac{\sqrt{2}}{2}$$

11)

$$\lim_{\delta x\to 0}\sum_{x=\pi/4}^{\pi/2} \frac{(\cos 2x+1)\operatorname{cosec} x}{\sin 2x}\,\delta x = \int_{\pi/4}^{\pi/2} \frac{(\cos 2x+1)\operatorname{cosec} x}{\sin 2x}\,dx$$
$$= \int_{\pi/4}^{\pi/2} \frac{(2\cos^2 x-1+1)\times\frac{1}{\sin x}}{2\sin x\cos x}\,dx$$
$$= \int_{\pi/4}^{\pi/2} \frac{2\cos^2 x}{2\sin^2 x\cos x}\,dx$$

$$= \int_{\pi/4}^{\pi/2} \frac{\cos x}{\sin^2 x}\,dx = \int_{\pi/4}^{\pi/2} \operatorname{cosec} x \cot x\,dx$$
$$= [-\operatorname{cosec} x]_{\pi/4}^{\pi/2}$$
$$= (-\operatorname{cosec} \tfrac{\pi}{2}) - (-\operatorname{cosec} \tfrac{\pi}{4})$$
$$= (-\tfrac{1}{\sin(\pi/2)}) - (-\tfrac{1}{\sin(\pi/4)})$$
$$= (-1) - (-\sqrt{2}) = \sqrt{2} - 1$$

12)

$$\int_{\pi/6}^{\pi/3} \frac{2\cos x}{\sin 2x(\operatorname{cosec} x - \sin x)}\,dx = \int_{\pi/6}^{\pi/3} \frac{2\cos x}{2\sin x\cos x(\frac{1}{\sin x} - \sin x)}\,dx$$
$$= \int_{\pi/6}^{\pi/3} \frac{1}{\sin x(\frac{1}{\sin x} - \sin x)}\,dx$$
$$= \int_{\pi/6}^{\pi/3} \frac{1}{1-\sin^2 x}\,dx = \int_{\pi/6}^{\pi/3} \frac{1}{\cos^2 x}\,dx$$
$$= \int_{\pi/6}^{\pi/3} \sec^2 x\,dx = [\tan x]_{\pi/6}^{\pi/3}$$
$$= (\tan \tfrac{\pi}{3}) - (\tan \tfrac{\pi}{6}) = (\sqrt{3}) - (\tfrac{\sqrt{3}}{3}) = \tfrac{2\sqrt{3}}{3}$$

13 a)

$$\sin 3x \equiv \sin(2x + x) \equiv \sin 2x\cos x + \cos 2x \sin x$$
$$\equiv (2\sin x\cos x)\cos x + (1 - 2\sin^2 x)\sin x$$
$$\equiv 2\sin x\cos^2 x + \sin x - 2\sin^3 x$$
$$\equiv 2\sin x(1 - \sin^2 x) + \sin x - 2\sin^3 x$$
$$\equiv 2\sin x - 2\sin^3 x + \sin x - 2\sin^3 x$$
$$\equiv 3\sin x - 4\sin^3 x$$

13 b)

$$\sin 3x = 3\sin x - 4\sin^3 x \Rightarrow \sin^3 x = \frac{3\sin x - \sin 3x}{4} = \tfrac{3}{4}\sin x - \tfrac{1}{4}\sin 3x$$
$$\int \sin^3 x\,dx = \int (\tfrac{3}{4}\sin x - \tfrac{1}{4}\sin 3x)\,dx = -\tfrac{3}{4}\cos x + \tfrac{1}{12}\cos 3x + c$$

14 a)

$$\cos 3x \equiv \cos(2x + x) \equiv \cos 2x\cos x - \sin 2x \sin x$$
$$\equiv (2\cos^2 x - 1)\cos x - 2\sin x\cos x \times \sin x$$
$$\equiv 2\cos^3 x - \cos x - 2\sin^2 x\cos x$$

$$\equiv 2\cos^3 x - \cos x - 2(1-\cos^2 x)\cos x$$
$$\equiv 2\cos^3 x - \cos x - 2\cos x + 2\cos^3 x$$
$$\equiv 4\cos^3 x - 3\cos x$$

14 b)

$$\cos 3x = 4\cos^3 x - 3\cos x \Rightarrow \cos^3 x = \frac{\cos 3x + 3\cos x}{4} = \tfrac{1}{4}\cos 3x + \tfrac{3}{4}\cos x$$
$$\int \cos^3 x\,dx = \int (\tfrac{1}{4}\cos 3x + \tfrac{3}{4}\cos x)\,dx = \tfrac{1}{12}\sin 3x + \tfrac{3}{4}\sin x + c$$

15)

$$\int_0^{1/2} \sqrt{\frac{16x}{1-x}}\,dx$$

$$x = \cos^2 u \Rightarrow \frac{dx}{du} = -2\sin u\cos u \Rightarrow dx = -2\sin u\cos u\,du$$

Change limits:

$$x = \tfrac{1}{2} \Rightarrow \tfrac{1}{2} = \cos^2 u \Rightarrow \cos u = \tfrac{1}{\sqrt{2}} \Rightarrow u = \arccos\tfrac{1}{\sqrt{2}} = \tfrac{\pi}{4}$$
$$x = 0 \Rightarrow 0 = \cos^2 u \Rightarrow \cos u = 0 \Rightarrow u = \tfrac{\pi}{2}$$

$$\begin{aligned}
\int_0^{1/2} \sqrt{\frac{16x}{1-x}}\,dx &= \int_{\pi/2}^{\pi/4} \sqrt{\frac{16\cos^2 u}{1-\cos^2 u}} \times -2\sin u\cos u\,du \\
&= \int_{\pi/4}^{\pi/2} \sqrt{\frac{16\cos^2 u}{\sin^2 u}} \times 2\sin u\cos u\,du \\
&= \int_{\pi/4}^{\pi/2} \frac{4\cos u}{\sin u} \times 2\sin u\cos u\,du \\
&= \int_{\pi/4}^{\pi/2} 8\cos^2 u\,du = \int_{\pi/4}^{\pi/2} 8 \times \tfrac{1}{2}(\cos 2u + 1)\,du \\
&= \int_{\pi/4}^{\pi/2} (4\cos 2u + 4)\,du = [2\sin 2u + 4u]_{\pi/4}^{\pi/2} \\
&= (2\sin(2\times\tfrac{\pi}{2}) + 4\times\tfrac{\pi}{2}) - (2\sin(2\times\tfrac{\pi}{4}) + 4\times\tfrac{\pi}{4}) \\
&= (0 + 2\pi) - (2 + \pi) = \pi - 2
\end{aligned}$$

16)

$$\int_{-1}^{5} \sqrt{(x+1)(5-x)}\,dx$$

$$x = 2 - 3\cos u \Rightarrow \frac{dx}{du} = 3\sin u \Rightarrow dx = 3\sin u\,du$$

Change limits:

$$x = 5 \Rightarrow 5 = 2 - 3\cos u \Rightarrow \cos u = -1 \Rightarrow u = \arccos(-1) = \pi$$

$$x = -1 \Rightarrow -1 = 2 - 3\cos u \Rightarrow \cos u = 1 \Rightarrow u = 0$$

$$\begin{aligned}
\int_{-1}^{5} \sqrt{(x+1)(5-x)}\,dx &= \int_{0}^{\pi} \sqrt{((2-3\cos u)+1)(5-(2-3\cos u))} \times 3\sin u\,du \\
&= 3\int_{0}^{\pi} \sqrt{(3-3\cos u)(3+3\cos u)}\sin u\,du \\
&= 3\int_{0}^{\pi} \sqrt{9-9\cos^2 u}\sin u\,du \\
&= 3\int_{0}^{\pi} \sqrt{9(1-\cos^2 u)}\sin u\,du \\
&= 3\int_{0}^{\pi} \sqrt{9\sin^2 u}\sin u\,du \\
&= \int_{0}^{\pi} 9\sin^2 u\,du = \int_{0}^{\pi} 9\times\tfrac{1}{2}(1-\cos 2u)\,du \\
&= \int_{0}^{\pi} (\tfrac{9}{2}-\tfrac{9}{2}\cos 2u)\,du = [\tfrac{9}{2}u-\tfrac{9}{4}\sin 2u]_0^{\pi} \\
&= (\tfrac{9}{2}\times\pi-\tfrac{9}{4}\sin(2\times\pi))-(\tfrac{9}{2}\times 0-\tfrac{9}{4}\sin(2\times 0)) \\
&= (\tfrac{9\pi}{2}-0)-(0-0) = \tfrac{9\pi}{2}
\end{aligned}$$